

2. A, B game tossing fair coin. If heads, A pays one unit to B. Conversely, B pays to A (1u) if tails. (2)

X_n = net gain A after n tosses

(a) $IP(X_3 = -1, X_5 = -3, X_9 = -3)$

$$\begin{aligned} & IP(X_9 = -3, X_5 = -3, X_3 = -1) \\ &= IP(X_9 = -3 | X_5 = -3, X_3 = -1) \cdot P(X_5 = -3, X_3 = -1) \\ &= IP(X_9 = -3 | X_5 = -3) \cdot P(X_5 = -3 | X_3 = -1) \cdot P(X_3 = -1) \\ &= IP(X_{9-5} = -3 - (-3)) \cdot P(X_{5-3} = -3 - (-1)) \cdot P(X_3 = -1) \\ &= IP(X_4 = 0) \cdot P(X_2 = -2) \cdot P(X_3 = -1) \\ &= \left[\binom{4}{2} \frac{1}{2^4} \right] \cdot \left(\frac{1}{2} \right)^2 \cdot \binom{3}{2} \frac{1}{2^3} \\ &= \frac{4 \cdot 3 \cdot 2}{2 \cdot 2} \cdot \frac{1}{2^4} \cdot \frac{1}{2^2} \cdot \frac{3!}{2!} \cdot \frac{1}{2^3} = \frac{3^2}{2^{4+1+3}} = \frac{9}{256} \end{aligned}$$

b) If $X_{50} = 0$, prob. that $IP(X_{53} = -1, X_{55} = -3, X_{59} = -3)$

Applying temporal stationarity, same result as a)

c) By CLT, approximate for $IP(-196 \leq X_{10000} \leq 196)$

$$\frac{X_n}{\sqrt{n}} \xrightarrow{d} N(0,1)$$

$$\begin{aligned} IP(-196 \leq X_{10000} \leq 196) &\approx F_{N(0,1)}\left(\frac{196}{\sqrt{10000}}\right) - F\left(\frac{-196}{\sqrt{10000}}\right) \\ &= 0.950 \end{aligned}$$

Using 4(a), # of trajcts. of SRW s.t. $X_{2m} > 0, Y_{2m} > 0$ (8)
 is $\left[IP(X_{2m} > 0, Y_{2m} > 0) \times \# \text{possible walks} \right] = \binom{2m}{m}$

On the other hand, # trajcts that conclude w/ eq. # votes for both candidates corresponds to the number of paths from $(0,0)$ to $(2m,0)$ that don't touch $y=-1$.

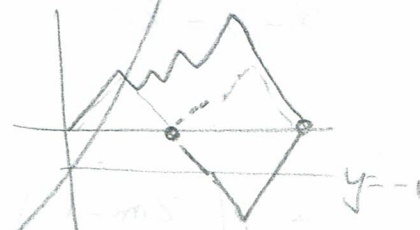
By reflection principle all possible paths that end in draw

$$\binom{2m}{m} - \binom{2m}{m+1} = \frac{1}{m+1} \binom{2m}{m}$$

bad paths

So

$$\frac{\frac{1}{m+1} \binom{2m}{m}}{\binom{2m}{m}} = \frac{1}{m+1}$$



A has the lead or
Same as B

7. A vehicle is in the plane at position $(0,0)$.

At every instant it moves w/ eq. probability either to the North or to the East. We know that at instant $2m$ the vehicle is at position (m,m) . Find the probability that it has not passed through any point of the form (i,i) for $0 < i < m$.

We identify North with $+1$ and East with -1 . So

$$X_n = \#N - \#E$$

A point in the diagonal (i,i) is equivalent to $\#N = \#E$, so

$$X_{2n} = 0.$$

Following the notation in the theory, we can compute the probability of the event as

$$IP(\text{doesn't pass through } (i,i), 0 < i < m \mid X_{2n} = 0) = \frac{IP(Z = 2m)}{IP(X_{2n} = 0)} \quad \textcircled{=}$$

$$IP(Z=2k) = \frac{1}{2^{k-1}} \cdot IP(X_{2k}=0)$$

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$$\Rightarrow \frac{IP(Z=2n)}{IP(X_{2n}=0)} = \frac{\frac{1}{2^{n-1}} \cdot IP(X_{2n}=0)}{IP(X_{2n}=0)} = \frac{1}{2^{n-1}}$$

8. At each time unit, a system is busy with prob $p = \frac{1}{2}$ and idle with prob $q = 1-p$. At time k , the performance level of the system, denoted as X_k , is defined as the difference between the number of busy times and the number of idle times.

a) What's the prob that the performance levels are 1, 2, and 3 ~~times~~ at times 5, 8, 15 respectively?

~~Performance level at:~~

~~$$\text{time 3} \rightarrow \binom{3}{2} \cdot p^2 \cdot q = 3 p^2 q = 3 \cdot \frac{1}{4} \cdot \frac{1}{2} = \frac{3}{8}$$~~

~~$$IP(X_5=1, X_8=2, X_{15}=3)$$~~

~~$$= IP(X_{15}=3 | X_8=2, X_5=1) \cdot IP(X_8=2, X_5=1)$$~~

~~$$= IP(X_{15}=3 | X_8=2) \cdot IP(X_8=2 | X_5=1) \cdot IP(X_5=1)$$~~

~~$$= IP(X_7=3 | X_0=2) \cdot IP(X_3=2 | X_0=1) \cdot IP(X_5=1)$$~~

~~$$= IP(X_7=1 | X_0=0) \cdot IP(X_3=1 | X_0=0) \cdot IP(X_5=1)$$~~

~~$$= IP(X_7=1) IP(X_3=1) \cdot IP(X_5=1)$$~~

~~$$= \binom{7}{4} p^7 q + \binom{3}{2} p^2 q + \binom{5}{3} p^4 q = \binom{7}{4} \binom{3}{2} \binom{5}{3} q^3 p^4$$~~

~~$$= 35 \cdot 105 \cdot \left(\frac{1}{2}\right)^{16}$$~~

$$\downarrow p=q=\frac{1}{2}$$

$$\frac{525}{16384}$$

~~$$= 105 \cdot \left(\frac{1}{2}\right)^{16}$$~~

(10)

h) What's the prob. that performance remains pos. for k consecutive time units if the performance starts at 1?

(Hint: Use that if $\{Z_n: n \geq 0\}$ is a SRW then

$$P(Z_1 \neq 0, \dots, Z_{2m} \neq 0) = P(Z_{2m} = 0)$$

$$P(X_1 \geq 0, X_2 \geq 0, \dots, X_k \geq 0 | X_0 = 1) = P(X_1 \geq -1, \dots, X_{2m} \geq -1 | X_0 = 0)$$

$$= P(X_1 \geq 0, \dots, X_{2m} \geq 0 | X_0 = 0) \stackrel{qa}{=} \binom{2m}{m} \frac{1}{2^{2m}}$$

9. In a ballot, candidate A receives m votes and candidate B receives n votes, where $m > n$

a) Prove that the number of voting sequences beginning with a vote for B is $\binom{m+n-1}{m}$

If the first vote goes for B, then we have $n-1$ votes left for B and m votes left for A. This is equivalent as counting the number of possible paths in a random walk of length $m+(n-1)$, where we get m votes for A

$$\binom{m+n-1}{m}$$

b) Use a reflection argument to show that any voting sequence that starts with a vote for A and results in a tie at some point corresponds one-to-one with a voting sequence that starts with a vote for B.

We represent each vote as a rw: $A \equiv +1$, $B \equiv -1$.

A sequence that starts with A is like $X_0 = 0$, $X_1 = 1$.

Suppose there's a tie at some point. Let T be

$$T = \min \{ k \geq 1 : X_k = 0 \}$$

the first instant of tie after the first vote.

(11)

Now, we pick a sequence such that

starts with A and touches 0 at some point.

And we transform it by reflecting the section from the beginning till the moment of first draw T:

$$A \leftrightarrow B$$

in the first T votes, and we leave the rest of the walk as it was. For example

$$AA \underset{\substack{\uparrow \\ T=4}}{BB} ABA \longrightarrow BB \ AA \ ABA$$

This transformation is invertible: if we start from B, pick the first instant of tie and reflect the initial section, we get the original sequence that started with A back.

So there's a bijection:

$\left. \begin{array}{l} \text{Sequences that start with A} \\ \text{and touch zero (tie) at} \\ \text{some point} \end{array} \right\} \longleftrightarrow \left. \begin{array}{l} \text{Sequences that start} \\ \text{with B} \end{array} \right\}$

c) From (a) and (b), deduce that the probability that candidate A was always ahead of candidate B during the vote counting is $(m-m)/(m+m)$.

The total number of possible paths is $\binom{m+m}{m}$.

The total number of paths where A is always ahead can be computed as the total number of paths minus the incorrect paths: the ones that start with A and touch zero at some point and the sequences that start with B. We know that both sets have the same cardinality by (b).

So the probability is

$$\frac{\binom{m+n}{m} - 2 \cdot \binom{m+n-1}{m}}{\binom{m+n}{m}} = \frac{\frac{(m+n)!}{m!n!} - 2 \cdot \frac{(m+n-1)!}{m!(n-1)!}}{\frac{(m+n)!}{m!n!}} =$$

$$= 1 - 2 \cdot \frac{n}{m+n} = \frac{m+n-2n}{m+n} = \frac{m-n}{m+n}$$

10. Let $\{X_n : n \geq 0\}$ be a symmetric r.w st $X_0 = 0$. Prove that if a is a positive integer then,

$$IP(X_1 \neq 0, \dots, X_{n-1} \neq 0, X_n = a) = \frac{a}{n} IP(X_n = a)$$

[Hint: consider trajectories from $(1,1)$ to (n,a) in the (n, X_n) -plane representation of the r.w]

$$\frac{a}{n} \cdot IP(X_n = a) = \frac{a}{n} \cdot \binom{n}{\frac{a+n}{2}} \rightarrow \begin{aligned} \alpha + \beta &= n \\ \alpha - \beta &= a \end{aligned} \rightarrow \begin{aligned} 2\alpha &= a+n \\ \alpha &= \frac{a+n}{2} \end{aligned}$$

$$= \frac{a}{n} \cdot \frac{n!}{\left(\frac{a+n}{2}\right)! \left(\frac{n-a}{2}\right)!}$$

$$IP(X_1 \neq 0, \dots, X_{n-1} \neq 0, X_n = a) \leftrightarrow$$

$N_{\text{paths}} = \# \text{ paths start with pos. and end at } a \quad (1,1) \rightarrow (n,a)$

$- \# \text{ paths start with pos. and touch axis}$

reflection with $(1,-1) \rightarrow (n,a)$

$$= \binom{n-1}{\frac{a+n}{2}-1} - \binom{n-1}{\frac{a+n}{2}} = \binom{n-1}{r-1} - \binom{n-1}{r}$$

$$= \frac{(n-1)!}{(r-1)!(n-r)!} - \frac{(n-1)!}{r!(n-r-1)!} = \frac{r}{n} \binom{n}{r} - \frac{n-r}{n} \binom{n}{r} = \frac{2r-n}{n} \binom{n}{r}$$

PROBABILITY AND RANDOM PROCESSES

6. MARKOV CHAINS

4. Consider a particle performing a random walk on a directed graph with vertex set $\{1, 2, 3, 4\}$ and adjacency matrix $A = (a_{ij})$

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

Here $a_{ij} = 1$ if there's a directed edge from node i to node j ; otherwise, $a_{ij} = 0$. If the particle is located at node i , it randomly

selects one of its adjacent neighbors (with equal probability) and moves to that node.

a) Analyse the random walk using a Markov chain framework. Identify the type of Markov chain represented.

Since every node is chosen uniformly among the neighbors, the transition matrix is

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 1/3 & 1/3 & 1/3 & 0 \end{pmatrix}$$

So the chain is finite, irreducible and aperiodic. It is aperiodic because, for instance, we can come back to state 2 in two steps:

$2 \rightarrow 1 \rightarrow 2$, and 3 steps: $2 \rightarrow 3 \rightarrow 4 \rightarrow 2$ and

$d_2 = \gcd(2, 3) = 1$. Since the chain is irreducible, all states have period 1.

b) Determine stationary distribution. Assess whether it also serves as a limiting distribution and provide justification. For each node i , $1 \leq i \leq 4$, compute the expected number of steps required for the particle to return to i .

(3)

Stationary distribution $\pi = (\pi_1, \pi_2, \pi_3, \pi_4)$ satisfies

$$\pi = \pi P$$

$$\sum_{i=1}^4 \pi_i = 1, \quad (A)$$

$$\pi = \pi P \Leftrightarrow \pi^T = P^T \pi^T$$

$$\begin{pmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \\ \pi_4 \end{pmatrix} = \begin{pmatrix} 0 & 1/2 & 0 & 1/3 \\ 1 & 0 & 1/2 & 1/3 \\ 0 & 1/2 & 0 & 1/3 \\ 0 & 0 & 1/2 & 0 \end{pmatrix} \begin{pmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \\ \pi_4 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 & 1/2 & 0 & 1/3 \\ 1 & -1 & 1/2 & 1/3 \\ 0 & 1/2 & -1 & 1/3 \\ 0 & 0 & 1/2 & -1 \end{pmatrix} \begin{pmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \\ \pi_4 \end{pmatrix} \rightarrow \text{Observe that } \pi_1 = \pi_3$$

We can remove eq 3 and add (A)

$$\begin{pmatrix} -1 & 1/2 & 0 & 1/3 \\ 1 & -1 & 1/2 & 1/3 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1/2 & -1 \end{pmatrix} \begin{pmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \\ \pi_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \rightarrow \text{Solve via calculator}$$

$$(\pi_1, \pi_2, \pi_3, \pi_4) = \left(\frac{6}{25}, \frac{10}{25}, \frac{6}{25}, \frac{3}{25} \right)$$

Since the Markov Chain is finite, irreducible and aperiodic:

$$\lim_{n \rightarrow \infty} P_{ij}^n = \pi_j \quad \text{so} \quad \lim_{n \rightarrow \infty} P^n = \begin{pmatrix} \frac{6}{25} & \frac{10}{25} & \frac{6}{25} & \frac{3}{25} \\ \frac{6}{25} & \frac{10}{25} & \frac{6}{25} & \frac{3}{25} \\ \frac{6}{25} & \frac{10}{25} & \frac{6}{25} & \frac{3}{25} \\ \frac{6}{25} & \frac{10}{25} & \frac{6}{25} & \frac{3}{25} \end{pmatrix}$$

it is also a limiting distribution.

The expected number of steps can be computed as $\mu_i = 1/\pi_i$ so

$$(\mu_1, \mu_2, \mu_3, \mu_4) = \left(\frac{25}{6}, \frac{25}{10} = \frac{5}{2}, \frac{25}{6}, \frac{25}{3} \right)$$

c) Suppose the random walk begins at mode 0, where $3 \leq i \leq 4$. Calculate the expected number of steps until the particle first visits either mode 1 or mode 2. Solve this problem using two approaches: first-step analysis and by converting the Markov chain into an absorbing chain.

(4)

1st step

Let $h_i = E(T_{\{1,2\}} | X_0 = i)$

Observe that $h_1 = h_2 = 0$

$$\left. \begin{aligned} h_3 &= 1 + \frac{1}{2}h_2 + \frac{1}{2}h_4 = 1 + \frac{1}{2}h_4 \\ h_4 &= 1 + \frac{1}{3}h_1 + \frac{1}{3}h_2 + \frac{1}{3}h_3 = 1 + \frac{1}{3}h_3 \end{aligned} \right\} \begin{aligned} h_3 - \frac{1}{2}h_4 &= 1 \\ -\frac{1}{3}h_3 + h_4 &= 1 \end{aligned}$$

$h_3 = \frac{9}{5}, \quad h_4 = \frac{8}{5}$

Absorbing chain

Convert P into P^* by setting states $\{1,2\}$ as absorbents

$P^* = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 1/3 & 1/3 & 1/3 & 0 \end{pmatrix}$, transient states are $T = \{3,4\}$
absorbent states are $A = \{1,2\}$

Rearranging into canonical: $P^* = \left(\begin{array}{c|c} Q & R \\ \hline 0 & I \end{array} \right) = \left(\begin{array}{cc|cc} 0 & 1/2 & 0 & 1/2 \\ 1/3 & 0 & 1/3 & 1/3 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right)$

The fundamental matrix N :

$$N = (I - Q)^{-1} = \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & 1/2 \\ 1/3 & 0 \end{pmatrix} \right)^{-1} = \begin{pmatrix} 6/5 & 3/5 \\ 2/5 & 6/5 \end{pmatrix}$$

Waiting time can be computed as $T = N \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 9/5 \\ 8/5 \end{pmatrix}$

so $E(T_{\{1,2\}} | X_0 = 3) = \frac{9}{5}$ $E(T_{\{1,2\}} | X_0 = 4) = \frac{8}{5}$

5. Consider two urns containing a total of four balls.

(5)

At each step, one of the four balls is randomly chosen and transferred from its current urn to the other urn. Define the states as the number of balls in the first urn.

a) Provide a justification that the process constitutes a Markov Chain and determine the corresponding transition probability matrix.

Let X_n be the number of balls in the first urn after n moves. So $S = \{0, 1, 2, 3, 4\}$. If $X_n = i$ there are i balls in the first urn and $4-i$ in the second. So

$$P_{i,i-1} = \frac{i}{4}, \quad P_{i,i+1} = \frac{4-i}{4}$$

The transition matrix would be

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1/4 & 0 & 3/4 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 3/4 & 0 & 1/4 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

It's Markov because X_{n+1} only depends on X_n and not the previous ones.

b) Determine the stationary distribution. Assess whether this stationary distribution also serves as a limiting distribution and provide justification for your answer.

In this case, we can use the detailed balance equations

$$\pi_i P_{ij} = \pi_j P_{ji} \rightarrow \pi_i P_{i,i+1} = \pi_{i+1} P_{i+1,i}$$

$$\pi_i \cdot \frac{4-i}{4} = \pi_{i+1} \cdot \frac{i+1}{4} \rightarrow \boxed{\pi_{i+1} = \frac{4-i}{i+1} \pi_i}$$

So we get

(6)

$$\pi_1 = 4\pi_0, \quad \pi_2 = \pi_1 \cdot \frac{3}{2} = 6\pi_0$$

$$\pi_3 = \frac{2}{3}\pi_2 = 4\pi_0 \quad \pi_4 = \frac{1}{4}\pi_3 = \pi_0$$

$$\text{Since } \pi_0 + \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$$

$$\pi_0 \cdot [1 + 4 + 6 + 4 + 1] = 1 \rightarrow \boxed{\pi_0 = \frac{1}{16}}$$

$$(\pi_0, \pi_1, \pi_2, \pi_3, \pi_4) = \left(\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16} \right)$$

The chain is irreducible, but it is not aperiodic. It has period 2, because in each step the number of balls' parity changes in the first urn:

$$0 \rightarrow 1, \quad 1 \rightarrow 0 \text{ or } 2, \quad 2 \rightarrow 1 \text{ or } 3, \dots$$

So π is the stationary distribution, but it's not the limit distribution.

- c) Assume initially that the first urn contains all four balls. Calculate the expected number of movements required until it contains all four balls again.

$X_0 = 4$, and we want to compute:

$$\mu_4 = \mathbb{E}(T_4 | X_0 = 4).$$

Since on a finite irreducible chain: $\pi_i = \frac{1}{\mu_i}$

$$\mu_4 = \frac{1}{\pi_4} = \frac{1}{1/16} = 16, \text{ therefore } \boxed{\mathbb{E}(T_4 | X_0 = 4) = 16}$$

(7)

d) Now, suppose that each urn initially contains two balls. Calculate the expected number of movements required until all four balls are located in a single urn.

Now $X_0 = 2$ and we want to reach either 0 or 4.

Let $h_i = E(T_{\{0,4\}} | X_0 = i)$

Observe that $h_0 = 0 = h_4$

$$\text{For } i=1 \rightarrow h_1 = 1 + \frac{1}{4}h_0 + \frac{3}{4}h_2 = 1 + \frac{3}{4}h_2$$

$$i=2 \rightarrow h_2 = 1 + \frac{1}{2}h_1 + \frac{1}{2}h_3$$

$$i=3 \rightarrow h_3 = 1 + \frac{3}{4}h_2 + \frac{1}{4}h_4 = 1 + \frac{3}{4}h_2$$

$$\left. \begin{array}{l} h_1 = 1 + \frac{3}{4}h_2 \\ h_2 = 1 + \frac{1}{2}h_1 + \frac{1}{2}h_3 \\ h_3 = 1 + \frac{3}{4}h_2 \end{array} \right\} \begin{array}{l} h_1 = h_3 \\ \rightarrow \end{array} \begin{array}{l} h_2 = 1 + h_1 \\ h_1 = 1 + \frac{3}{4}h_2 \end{array}$$

$$h_2 = 1 + \frac{3}{4}h_2 + 1 \Rightarrow \frac{1}{4}h_2 = 2 \rightarrow h_2 = 8$$

$$\boxed{h_2 = E(T_{\{0,4\}} | X_0 = 2) = 8}$$

10. A vending machine dispenses items for one euro and is equipped with a box, denoted B, with a capacity of $2k$ one-euro coins. Initially box B contains k coins. (8)

Each customer independently pays with a one-euro coin with prob. p ($0 < p < 1$) depositing the coin into box B, or with a two-euro coin with prob. $q = 1 - p$, depositing the coin into separate box B'. In the latter case, the machine dispenses one euro from box B as change. When box B becomes empty, an employee refills it with k one-euro coins. When box B becomes full, the employee removes k one-euro coins.

Let X_n denote the number of coins in box B after n individuals (including both customers and employees) have interacted with the machine.

a) Give the transition probabilities of the Markov chain $\{X_n: n \geq 0\}$ and classify its states.

$X_n = \# \text{ one-euro coins at B}$

B has capacity $2k$, so the states are $S = \{0, 1, \dots, 2k\}$

For $1 \leq i \leq 2k-1$ $P_{i,i+1} = p$ because the client pays with a one-euro coin with prob. p . And

$P_{i,i-1} = q = 1 - p$ when client pays with 2-euro coin.

Also, $P_{0,k} = 1$ $P_{2k,k} = 1$

The chain is finite and irreducible because from any state you can reach any other state with $+1, -1$ movements or the filling/emptying moves $0 \rightarrow k, 2k \rightarrow k$.

h) $k=2$. Determine the stationary. What is the expected number of individuals who have interacted with the machine until box B again contains two coins? Compute $\lim_{n \rightarrow \infty} P(X_n = k)$ if this limit exists and justify the answer.

$$k=2 \rightarrow S = \{0, 1, 2, 3, 4\}$$

$$P = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ q & 0 & p & 0 & 0 \\ 0 & q & 0 & p & 0 \\ 0 & 0 & q & 0 & p \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix} \quad \text{and we look for } \pi = \pi P$$

$$\sum_{i=0}^4 \pi_i = 1$$

The equations are: $\pi_0 = q \pi_1$

$$\pi_1 = q \pi_2$$

$$\pi_2 = \pi_0 + p \pi_1 + q \pi_3 + \pi_4$$

$$\pi_3 = p \pi_2$$

$$\pi_4 = p \pi_3$$

If we express all in terms of π_2

$$\pi_0 = q^2 \pi_2, \pi_1 = q \pi_2, \pi_2 = \pi_2, \pi_3 = p \pi_2, \pi_4 = p^2 \pi_2$$

$$\sum_{i=0}^4 \pi_i = \pi_2 (q^2 + q + 1 + p + p^2) \stackrel{p+q=1}{=} \pi_2 (2 + p^2 + q^2)$$

$$= \pi_2 (2 + p^2 + q^2) = 1$$

$$\rightarrow \pi_2 = \frac{1}{2 + p^2 + q^2} \rightarrow (\pi_0, \pi_1, \pi_2, \pi_3, \pi_4) = \frac{(q^2, q, 1, p, p^2)}{2 + p^2 + q^2}$$

The chain is aperiodic, because I can come back to 2 in two steps ($2 \rightarrow 1 \rightarrow 2$) and three steps ($2 \rightarrow 3 \rightarrow 4 \rightarrow 2$), so

$d_2 = \gcd(2, 3) = 1$. Since the chain is finite, irreducible and aperiodic

$$\lim_{n \rightarrow \infty} P(X_n = 2) = \pi_2 = \frac{1}{2 + p^2 + q^2} = \frac{1}{2p^2 - 2p + 3}$$

c) let $p=1/2$. Determine the expected number of customers served before the machine becomes empty or full for the first time. [Hint: apply first step analysis to verify that the service time, starting with i coins in B, is $i(2k-i)$]. (10)

Before an employee appears, we kind of have a random walk (a symmetric random walk) in the states $\{0, 1, 2, \dots, 2k\}$ with barriers at 0 and $2k$.

Let $h_i = E(T_{\{0, 2k\}} | X_0 = i)$

Observe that $h_0 = h_{2k} = 0$.

For $1 \leq i \leq 2k-1$

$$h_i = 1 + \frac{1}{2} h_{i-1} + \frac{1}{2} h_{i+1} \quad (A)$$

The box at the beginning has k coins, $X_0 = k$

So, we rewrite (A)

$$2h_i = 2 + h_{i-1} + h_{i+1} \rightarrow h_{i+1} - 2h_i + h_{i-1} = -2$$

This is a second-order difference expression, we aim for a quadratic expression $h_i = ai^2 + bi + c$

$$a(i+1)^2 + b(i+1) + c - 2ai^2 - 2bi - 2c + a(i-1)^2 + b(i-1) + c = -2$$

$$a(i^2 + 1 + 2i + i^2 + 1 - 2i - 2i^2) = 2a = -2 \rightarrow a = -1$$

$$\text{Since } h_0 = 0 = (-1) \cdot 0^2 + b \cdot 0 + c \rightarrow c = 0$$

$$\text{Since } h_{2k} = 0 = (-1) \cdot (2k)^2 + b \cdot 2k = 0 \rightarrow b \cdot 2k = (2k)^2$$

$$b = 2k$$

$\rightarrow h_i = -i^2 + 2ki = i(2k-i)$ ↙ The given expression
Since we have proved it
there's no need to verify.

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Since we start at k , $x_0 = k$

$$h_n = k \cdot (2k - k) = k^2$$